

enabling you to build appropriate counter measures. Triggers can also be configured based on alerts; for example, a webhook call can be triggered to restart a service if your service is not giving response. The benefits of log analysis are:

- Reduces the time in troubleshooting and resolving issues in your systems.
 - Improves the user experience and reduces customer churn.
 - Helps in efficient use of IT resources.
 - Improves security and compliance of IT systems.
- Use cases of log analysis include:
- Troubleshooting issues in systems, networks and other IT resources.
 - Meeting security policies and audits, and regulatory compliance.
 - Understanding the behaviour of users in your systems.
 - Understanding and responding early to any security incidents and data breaches.
 - Rapid and early detection of failed processes.
 - Improving the performance of IT resources.

Let's now perform a log analysis with ELK Stack.

As stated earlier, ELK Stack is the acronym for three open source tools: Elasticsearch, Logstash and Kibana. Combining these three tools will provide you all the features required to perform log analysis.

1. **Elasticsearch** is a search and analytics engine. It stores and analyses the logs, security related events and metrics. It offers speed

and flexibility to handle this data with the use of indexes.

2. **Logstash** is a tool for shipping, processing and storing the logs collected from different sources. It is a data collection engine with capabilities of real-time pipelining. It collects data from different sources, normalises it, and puts it in a stash called Elasticsearch.
3. **Kibana** is a user interface to visualise Elasticsearch data, and to create charts and graphs for efficient log analysis. Alerts can also be set in case of any issue.

Figure 1 gives the basic architectural flow of these three components for log analysis.

In this figure, you can view the architectural flow as follows:

1. All the logs from various IT resources are first sent to the Log Collector.
2. In ELK, the log collection is different kinds of beats like heartbeat, filebeat, metricbeat, etc. These beats are configured based on which kind of data you want to check the status of for any service (heartbeat), getting logs from various log files (filebeat) or getting different metrics (metricbeat). These beats collect and send all these logs to Logstash.
3. Once the data is with Logstash, it aggregates and processes it, and stashes it to Elasticsearch for analysis.
4. Elasticsearch stores and analyses this data, and feeds the analysed data to the Kibana dashboard.

5. In the last step, Kibana visualises this data in some form of graph or chart by querying it as required by the stakeholders.

This is just the basic simple flow configured for learning purpose. But for production grade implementation, adding components like Redis, Kafka and RabbitMQ for resiliency is recommended.

Configuring ELK for log analysis

Considering the ELK Stack is already installed, we can directly configure log analysis to capture nginx server logs from a Linux system. You can use this process for Apache logs as well. In a Linux based system, server logs are present under the `/var/log` directory and nginx logs are in the `/var/log/nginx` directory.

Follow the steps given below to configure nginx logs in ELK.

1. First, install filebeat:

```
> curl -L -O https://artifacts.elastic.co/downloads/beats/filebeat/filebeat-7.12.0-linux-x86_64.tar.gz
```

```
> tar xzvf filebeat-7.12.0-linux-x86_64.tar.gz
```

2. Update the filebeat configuration file at location in `<filebeat-directory>/config/filebeat.yml`. Edit the Elasticsearch details as given below:

```
Config
output.elasticsearch:
  hosts: ["your-elasticsearch-host:9200"]
  username: "YOUR_ELASTICSEARCH_USERNAME"
  password: "YOUR_ELASTICSEARCH_PASSWORD"
```

Also, update the Kibana details as given below:

```
setup.kibana:
  host: "your-kibana-host:5601"
```

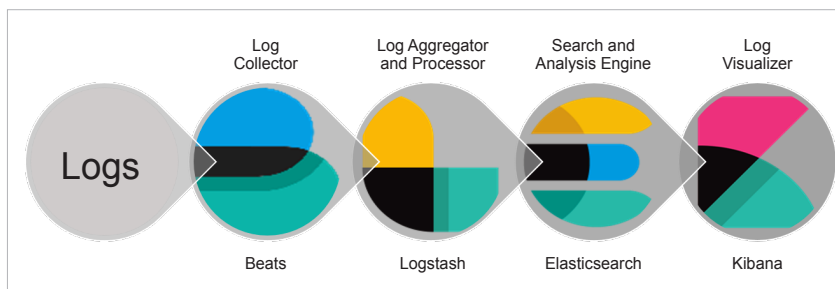


Figure 1: Basic architectural flow of ELK Stack